

Qualification Pack



Water Pump Operator

QP Code: PSC/Q0118

Version: 2.0

NSQF Level: 4

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower
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PSC/Q0118: Water Pump Operator

Brief Job Description

The Water Pump Operator is responsible for the installation, operation and maintenance of pumps. The individual will also operate and maintain the water supply, water distribution and waste water treatment systems.

Personal Attributes

The individual must be able to work independently and be comfortable in performing laborious work. The person should be good at following instructions and a team player with a positive attitude.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSC/N0151: Install Pump Systems and Related Equipment](#)
2. [PSC/N0152: Operate and Maintain Pumps and Related Machinery](#)
3. [PSC/N0153: Perform Maintenance of Local Ground Water Source and Water Quality Monitoring at Water Supply Stations](#)
4. [PSC/N0154: Maintain of Water Distribution and Storage Systems at Water Supply Stations](#)
5. [PSC/N0155: Operate and Maintain Water Treatment Units](#)
6. [PSC/N0136: Apply Health and Safety Practices at the Workplace](#)
7. [PSC/N0137: Work Effectively with Others](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
Country	India

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NSQF Level	4
Credits	14
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7126.0101
Minimum Educational Qualification & Experience	12th grade Pass OR Pursuing 3rd year of 3-year diploma after 10th OR 10th grade pass with 1 year NTC plus CITS (or NAC) OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS OR Previous relevant Qualification of NSQF Level (3 as Assistant Plumber - General with 8th Grade Pass) with 2 Years of experience Relevant
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	24/02/2025
NSQC Approval Date	24/02/2022
Version	2.0
Reference code on NQR	2022/PLUM/WMPs/05465
NQR Version	1

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PSC/N0151: Install Pump Systems and Related Equipment

Description

This unit is about installing pumps systems and related equipment for different plumbing applications in residential, agricultural and commercial setups.

Scope

The scope covers the following :

- Prepare for work
- Install pump and related equipment
- Check the installation

Elements and Performance Criteria

Prepare for work

To be competent, the user/individual on the job must be able to:

- PC1.** assemble pump components and equipment
- PC2.** prepare the tools, area and materials for the task
- PC3.** locate and mark position for inlet and outlet supply connections of pump

Install pump and related equipment

To be competent, the user/individual on the job must be able to:

- PC4.** fix the pump at the designated location as per instruction
- PC5.** connect the hoses of inlet and outlet supply to the pump
- PC6.** make provisions for electrical and other required connections
- PC7.** ensure prevention of any contact of water and electrical connections with each other
- PC8.** install and connect pump components without any damage to pump, fixture, pipe work, the surrounding environment, or to other services
- PC9.** adjust pressure/flow as per required supply and demand

Check the installation

To be competent, the user/individual on the job must be able to:

- PC10.** check installed pump systems for correct functioning and compliance with specifications
- PC11.** check for cracks, defects and anomalies in the pumping apparatus
- PC12.** check for condition of couplings in the equipment and pumping on both suction and discharge sides
- PC13.** check the oil level, fuel level, radiator coolant and engine condition of a diesel operated pump.
- PC14.** check air release valve, and prime the pump

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** standards, policies, and procedures followed in the company relevant to employment and performance conditions
- KU2.** workplace safety requirements, hazard reporting and handling procedures
- KU3.** various types of pumps used in residential, agricultural and commercial setups
- KU4.** terminologies used in pump systems
- KU5.** components of pump systems and related equipment
- KU6.** tools and materials used in pump installation
- KU7.** preparatory requirements for pump system installation
- KU8.** types of water supply and its implication on pumping system
- KU9.** performance measures, applications and properties of water pumps
- KU10.** manufacturers' specifications related to the installation and fitting of pumps
- KU11.** units of measurements
- KU12.** how to calculate material requirements
- KU13.** procedure of installing water pumps
- KU14.** levelling and alignment procedures
- KU15.** material disposal procedure and its importance
- KU16.** basic fittings (valves, clamps, elbows, etc.) in the pumping apparatus
- KU17.** different types of pumping apparatus (reciprocating, rotary etc.) and associated equipment and purposes
- KU18.** gauges, dials, monitoring apparatus and their purpose
- KU19.** condition monitoring of the equipment
- KU20.** dos and don'ts of various pumping systems as per their Standard Operating Procedures (SOPs) specified by the equipment manufacturer.
- KU21.** different types of valves and their functioning (stop valve, non-return valve, etc.)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS3.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS4.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS5.** plan one's daily tasks to achieve maximum productivity
- GS6.** establish priorities and deadlines in consultation with others and record them
- GS7.** be punctual and work as per agreed priorities
- GS8.** manage distractions and maintain workplace discipline
- GS9.** listen to customer's concerns and doubts carefully and address them
- GS10.** be courteous
- GS11.** identify ways to increase productivity and reduce errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for work</i>	10	14	3	3
PC1. assemble pump components and equipment	4	5	1	1
PC2. prepare the tools, area and materials for the task	3	4	1	1
PC3. locate and mark position for inlet and outlet supply connections of pump	3	5	1	1
<i>Install pump and related equipment</i>	10	21	7	3
PC4. fix the pump at the designated location as per instruction	2	8	4	1
PC5. connect the hoses of inlet and outlet supply to the pump	2	5	1	0.5
PC6. make provisions for electrical and other required connections	1	2	0.5	0.5
PC7. ensure prevention of any contact of water and electrical connections with each other	1	2	0.5	0.5
PC8. install and connect pump components without any damage to pump, fixture, pipe work, the surrounding environment, or to other services	3	2	0.5	-
PC9. adjust pressure/flow as per required supply and demand	1	2	0.5	0.5
<i>Check the installation</i>	10	14	4	1
PC10. check installed pump systems for correct functioning and compliance with specifications	2	4	2	1
PC11. check for cracks, defects and anomalies in the pumping apparatus	2	2.5	0.5	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check for condition of couplings in the equipment and pumping on both suction and discharge sides	2	2.5	0.5	-
PC13. check the oil level, fuel level, radiator coolant and engine condition of a diesel operated pump.	2	2.5	0.5	-
PC14. check air release valve, and prime the pump	2	2.5	0.5	-
NOS Total	30	49	14	7

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0151
NOS Name	Install Pump Systems and Related Equipment
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

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PSC/N0152: Operate and Maintain Pumps and Related Machinery

Description

This unit is about efficient pump operation and preventive maintenance of the pump and related machinery such as motor, valves and electrical components, etc.

Scope

The scope covers the following :

- Operate the pump
- Maintain the pump and related machinery

Elements and Performance Criteria

Operate the pump

To be competent, the user/individual on the job must be able to:

- PC1.** prepare a schedule for running the main pumps and the standby
- PC2.** prime the centrifugal pump before start of operation
- PC3.** ensure operation of the pump with full efficiency to improve output gain
- PC4.** operate the delivery valve effectively to avoid overloading and sudden change of water pressure
- PC5.** close bypass valves of reflux valve, sluice valve and butterfly valve during normal operation of the pumps
- PC6.** monitor sound, vibration, temperature and other related parameters to ensure that the pump is operating smoothly
- PC7.** maintain input/output parameters for the pump as per requirements
- PC8.** check pump motor alignment
- PC9.** take defined action in case of overload/under load when the load parameters are above/below the specified limits
- PC10.** maintain a record of all pump operation timings, voltage, current, reading on gauges and flow meter, temperature, water level and any problem that occurred during the operation

Maintain the pump and related machinery

To be competent, the user/individual on the job must be able to:

- PC11.** prepare and apply a monthly, quarterly, bi-annual and annual preventive maintenance schedule
- PC12.** clean the pump, related machinery, pump chamber filter beds, other structures and the area
- PC13.** inspect the pump, related machinery and water supply system for faults, leaks, sparks, improper functioning, wear and tear or evidence of tampering
- PC14.** record and report the faults, leaks, sparks, improper functioning, water supply complaints and tampering
- PC15.** apply appropriate oil where oil levels are low as per standard
- PC16.** repair or replace sparking motor and leaking or worn out parts

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- PC17.** check and rectify alignment of pump and drive
- PC18.** tighten the foundation bolts and holding down bolts of pump and motor mounting
- PC19.** calibrate all vital instruments such as pressure gauge, vacuum gauge, ammeter, voltmeter, watt meters, frequency meter, tachometer and flow meter
- PC20.** conduct performance test of the pump for discharge, head and efficiency
- PC21.** troubleshoot faults as per standard operating procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to employment and performance conditions
- KU2.** workplace safety requirements, hazard reporting and handling procedures
- KU3.** basic pumping / operation terminology
- KU4.** different types of operation and maintenance schedules
- KU5.** types of pump to be operated
- KU6.** importance of preparing and following a schedule for pump operation and maintenance
- KU7.** factors to be considered for preparing an operation and maintenance schedule
- KU8.** need for standby pumps
- KU9.** how to improve output gain from the pump
- KU10.** need and process for avoiding overloading and sudden change of water pressure
- KU11.** function and operation of various valves such as bypass valves of reflux valve, sluice valve and butterfly valve during operation of the pumps
- KU12.** factors that effect the efficiency of the pump such as voltage, current, location, starting load, etc.
- KU13.** common sounds, vibrations, temperature and other related parameters that can change during the operation of the pump and their significance.
- KU14.** impact of various physical parameters like temperature, pressure, etc on the properties of final output
- KU15.** various input/output parameters of the pump for various types of requirements
- KU16.** purpose and procedure for checking pump motor alignment
- KU17.** normal load parameters and indicators of overload/underload
- KU18.** action to be taken in case of overload/underload
- KU19.** importance of maintaining a record of pump operation timings, voltage, current, reading on gauges and flow meter, temperature, water level and any problem that occurred during the operation
- KU20.** importance of preventive maintenance
- KU21.** various preventive maintenance tasks to be performed daily, monthly, quarterly, bi-annually and annually
- KU22.** considerations to be taken while making a preventive maintenance schedule
- KU23.** dos and don'ts while cleaning the pump, related machinery, pump chamber filter beds, other structures and the area

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- KU24.** importance of regular inspections of the pump, related machinery and water supply system
- KU25.** common indicators of faults, leaks, sparks, improper functioning, wear and tear or tampering in a pump and related machinery
- KU26.** importance of timely recording and reporting of faults, leaks, sparks, improper functioning, water supply complaints and tampering
- KU27.** appropriate oil levels and procedure to check it
- KU28.** action to be taken if motor is sparking or parts are leaking or worn out
- KU29.** procedure to check and rectify alignment of pump and drive
- KU30.** various bolts to be tightened
- KU31.** calibration procedure of all vital instruments such as pressure gauge, vacuum gauge, ammeter, voltmeter, watt meters, frequency meter, tachometer and flow meter
- KU32.** performance testing procedure of the pump for discharge, head and efficiency
- KU33.** action to be taken for common faults

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** measure all dimensions in metric scale
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS5.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS6.** plan one's daily tasks to achieve maximum productivity
- GS7.** be punctual and work as per agreed priorities
- GS8.** listen to customer's concerns and doubts carefully and address them
- GS9.** be courteous and polite with customers and team members
- GS10.** establish workable solutions for problems in hand in consultation with others and record them
- GS11.** breakdown relevant work process into its constituent activities for ease of analysis
- GS12.** identify ways to increase productivity and reduce error

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the pump</i>	13	23	5	3
PC1. prepare a schedule for running the main pumps and the standby	1	2	0.5	0.5
PC2. prime the centrifugal pump before start of operation	2	3	0.5	-
PC3. ensure operation of the pump with full efficiency to improve output gain	1	2	-	0.5
PC4. operate the delivery valve effectively to avoid overloading and sudden change of water pressure	1	2	0.5	-
PC5. close bypass valves of reflux valve, sluice valve and butterfly valve during normal operation of the pumps	1	2	0.5	-
PC6. monitor sound, vibration, temperature and other related parameters to ensure that the pump is operating smoothly	2	3	1	0.5
PC7. maintain input/output parameters for the pump as per requirements	2	3	1	0.5
PC8. check pump motor alignment	1	2	0.5	0.5
PC9. take defined action in case of overload/under load when the load parameters are above/below the specified limits	1	2	-	0.5
PC10. maintain a record of all pump operation timings, voltage, current, reading on gauges and flow meter, temperature, water level and any problem that occurred during the operation	1	2	0.5	-
<i>Maintain the pump and related machinery</i>	17	26	9	4
PC11. prepare and apply a monthly, quarterly, bi-annual and annual preventive maintenance schedule	1	2	0.5	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. clean the pump, related machinery, pump chamber filter beds, other structures and the area	1	2	1	0.5
PC13. inspect the pump, related machinery and water supply system for faults, leaks, sparks, improper functioning, wear and tear or evidence of tampering	3	4	1	1
PC14. record and report the faults, leaks, sparks, improper functioning, water supply complaints and tampering	1	1	0.5	-
PC15. apply appropriate oil where oil levels are low as per standard	1	2	0.5	0.5
PC16. repair or replace sparking motor and leaking or worn out parts	2	3	1	-
PC17. check and rectify alignment of pump and drive	1	2	0.5	0.5
PC18. tighten the foundation bolts and holding down bolts of pump and motor mounting	1	2	1	-
PC19. calibrate all vital instruments such as pressure gauge, vacuum gauge, ammeter, voltmeter, watt meters, frequency meter, tachometer and flow meter	2	2	1	-
PC20. conduct performance test of the pump for discharge, head and efficiency	2	3	1	0.5
PC21. troubleshoot faults as per standard operating procedure	2	3	1	0.5
NOS Total	30	49	14	7

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0152
NOS Name	Operate and Maintain Pumps and Related Machinery
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

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PSC/N0153: Perform Maintenance of Local Ground Water Source and Water Quality Monitoring at Water Supply Stations

Description

This unit deals in detail with the maintenance of local ground water source, which include, dug well, hand pump and tube/bore well and monitoring of water quality at water supply stations.

Scope

The scope covers the following :

- Maintain dug well
- Maintain hand pump
- Maintain tube well and bore well
- Monitor water quality
- Maintain spring source and surface water source

Elements and Performance Criteria

Maintain dug well

To be competent, the user/individual on the job must be able to:

- PC1.** clean the concrete apron and debris in the well
- PC2.** check the concrete apron and well seal for cracks and repair with cement mortar as needed
- PC3.** check the condition of the rope, pulley, support posts, bucket and fence and repair or replace when needed.
- PC4.** lubricate the pulley as needed with grease
- PC5.** record the water level with a rope scale
- PC6.** repair any structural damage to the well and surrounding structure
- PC7.** dewater the well and clean the bottom
- PC8.** inspect the well walls and lining and repair as needed
- PC9.** check the water level and deepen/de-silt the well especially in monsoon

Maintain hand pump

To be competent, the user/individual on the job must be able to:

- PC10.** check all the flange nuts and bolts, axle bolt, flange bolts and tighten as needed
- PC11.** tighten the handle axle nut and lock nut
- PC12.** verify whether hand pump is firm on its base and fix it if needed
- PC13.** open the cover and clean inside the pump
- PC14.** dismantle the hand pump for inspection/cleaning and reassemble after inspection
- PC15.** check the chain anchor bolt for proper position and tighten if needed
- PC16.** verify rusty patches, clean with a wire brush and apply anticorrosive paint
- PC17.** verify the discharge of water
- PC18.** verify the handle position and repair if needed

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- PC19.** verify whether guide bush, roller chain is not excessively worn out and replace if needed
- PC20.** clean and de-calcify pump components
- PC21.** take action as per standard operating procedure to troubleshoot faults occurring in the hand pump

Maintain tube well and bore well

To be competent, the user/individual on the job must be able to:

- PC22.** operate pump starter and isolation valve
- PC23.** check if readings on ammeter and voltmeter are normal – stop pump if electric motor is drawing too much current
- PC24.** verify whether adequate water is being delivered
- PC25.** clean the pump house
- PC26.** check for leaks in the rising main
- PC27.** remove the pump and rising main from the well and inspect
- PC28.** inspect pipes, electric cables, insulation between cables
- PC29.** record servicing and maintenance in logbook
- PC30.** re-cut corroded or damaged threads and replace badly corroded pipes
- PC31.** de-silt borehole if required
- PC32.** take action as per standard operating procedure to troubleshoot faults occurring in the tube or bore well

Maintain spring source and surface water source

To be competent, the user/individual on the job must be able to:

- PC33.** check whether the area is free from waste disposal and defecation
- PC34.** check there is no unwanted human intervention or animal intrusion
- PC35.** check intake for clogging and submergence
- PC36.** repair any small damages to the intake system
- PC37.** dewater and clean the bottom of the spring source
- PC38.** check the water level of the surface water source and de-silt as per requirement

Monitor water quality

To be competent, the user/individual on the job must be able to:

- PC39.** adhere to water sampling schedule as per standard operating procedure
- PC40.** identify a water sampling site representative to the source and in line with sampling guidelines
- PC41.** obtain water samples from dug well, hand pump, tube well and bore well for quality testing
- PC42.** test water quality using a field test kit
- PC43.** store and send samples for laboratory testing
- PC44.** interpret relevant information from water quality report
- PC45.** share water quality test results with appropriate authorities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** standards, policies, and procedures followed in the company relevant to the task
- KU2.** workplace safety requirements, hazard reporting and handling procedures
- KU3.** various ground water sources
- KU4.** structure and use of dug well, hand pump, tube well and bore well
- KU5.** maintenance activities performed for a dug well, hand pump, tube well and bore well and their importance
- KU6.** maintenance activities to be performed for spring source and surface water source and their importance
- KU7.** importance of water quality monitoring
- KU8.** water quality monitoring tasks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** measure all dimensions used in plumbing in metric scale
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS5.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS6.** plan one's daily tasks to achieve maximum productivity
- GS7.** be punctual and work as per agreed priorities
- GS8.** listen to customer's concerns and doubts carefully and address them
- GS9.** be courteous and polite with customers and team members
- GS10.** establish workable solutions for problems in hand in consultation with others and record them
- GS11.** breakdown relevant work process into its constituent activities for ease of analysis
- GS12.** identify ways to increase productivity and reduce errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain dug well</i>	6	9	3	1
PC1. clean the concrete apron and debris in the well	0.5	1	0.5	-
PC2. check the concrete apron and well seal for cracks and repair with cement mortar as needed	0.5	1	-	-
PC3. check the condition of the rope, pulley, support posts, bucket and fence and repair or replace when needed.	0.5	1	0.5	-
PC4. lubricate the pulley as needed with grease	0.5	1	0.5	-
PC5. record the water level with a rope scale	0.5	1	0.5	-
PC6. repair any structural damage to the well and surrounding structure	1	1	-	0.5
PC7. dewater the well and clean the bottom	0.5	1	0.5	-
PC8. inspect the well walls and lining and repair as needed	1	1	-	0.5
PC9. check the water level and deepen/de-silt the well especially in monsoon	1	1	0.5	-
<i>Maintain hand pump</i>	6	10	3	1
PC10. check all the flange nuts and bolts, axle bolt, flange bolts and tighten as needed	-	0.5	-	-
PC11. tighten the handle axle nut and lock nut	-	0.5	-	-
PC12. verify whether hand pump is firm on its base and fix it if needed	0.5	0.5	-	-
PC13. open the cover and clean inside the pump	0.5	1	0.5	-
PC14. dismantle the hand pump for inspection/cleaning and reassemble after inspection	0.5	2	1	-
PC15. check the chain anchor bolt for proper position and tighten if needed	0.5	0.5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. verify rusty patches, clean with a wire brush and apply anticorrosive paint	0.5	1	-	-
PC17. verify the discharge of water	0.5	0.5	-	-
PC18. verify the handle position and repair if needed	0.5	1	-	-
PC19. verify whether guide bush, roller chain is not excessively worn out and replace if needed	0.5	1	-	-
PC20. clean and de-calcify pump components	0.5	0.5	0.5	-
PC21. take action as per standard operating procedure to troubleshoot faults occurring in the hand pump	1.5	1	1	1
<i>Maintain tube well and bore well</i>	6.5	10	3	1
PC22. operate pump starter and isolation valve	0.5	0.5	0.5	-
PC23. check if readings on ammeter and voltmeter are normal – stop pump if electric motor is drawing too much current	1	1	0.5	0.5
PC24. verify whether adequate water is being delivered	0.5	0.5	0.5	-
PC25. clean the pump house	0.5	0.5	-	-
PC26. check for leaks in the rising main	0.5	1	-	-
PC27. remove the pump and rising main from the well and inspect	0.5	1.5	0.5	-
PC28. inspect pipes, electric cables, insulation between cables	0.5	1	-	0.5
PC29. record servicing and maintenance in logbook	0.5	0.5	-	-
PC30. re-cut corroded or damaged threads and replace badly corroded pipes	0.5	1	-	-
PC31. de-silt borehole if required	0.5	1	0.5	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC32. take action as per standard operating procedure to troubleshoot faults occurring in the tube or bore well	1	1.5	0.5	-
<i>Maintain spring source and surface water source</i>	6	8	3	2
PC33. check whether the area is free from waste disposal and defecation	1	1	0.5	0.5
PC34. check there is no unwanted human intervention or animal intrusion	1	1	0.5	0.5
PC35. check intake for clogging and submergence	1	1	0.5	0.5
PC36. repair any small damages to the intake system	1	1	0.5	0.5
PC37. dewater and clean the bottom of the spring source	1	2	0.5	-
PC38. check the water level of the surface water source and de-silt as per requirement	1	2	0.5	-
<i>Monitor water quality</i>	6.5	10	3	2
PC39. adhere to water sampling schedule as per standard operating procedure	0.5	1	-	0.5
PC40. identify a water sampling site representative to the source and in line with sampling guidelines	1	2	0.5	0.5
PC41. obtain water samples from dug well, hand pump, tube well and bore well for quality testing	1	2	0.5	-
PC42. test water quality using a field test kit	1	2	0.5	0.5
PC43. store and send samples for laboratory testing	1	1	0.5	-
PC44. interpret relevant information from water quality report	1	1	0.5	0.5
PC45. share water quality test results with appropriate authorities	1	1	0.5	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	31	47	15	7

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0153
NOS Name	Perform Maintenance of Local Ground Water Source and Water Quality Monitoring at Water Supply Stations
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

PSC/N0154: Maintain of Water Distribution and Storage Systems at Water Supply Stations

Description

This unit deals in detail with operation and maintenance of the water distribution and storage systems at water supply stations.

Scope

The scope covers the following :

- Maintain sump and storage tanks
- Maintain water meters
- Monitor and maintain flow meters
- Operate and maintain and repair pipelines

Elements and Performance Criteria

Maintain sump and storage tanks

To be competent, the user/individual on the job must be able to:

- PC1.** dewater sump and tank
- PC2.** clean, disinfect and rinse the sump and tank
- PC3.** desilt the area and pump house.
- PC4.** perform leakage test and carry out rectification if needed with epoxy coating, cement concreting, painting etc.

Maintain water meters

To be competent, the user/individual on the job must be able to:

- PC5.** clean the dirt box or strainer
- PC6.** replace gaskets upon its wear and tear
- PC7.** clean the chamber where meter is installed
- PC8.** prevent water seepage into the water meter
- PC9.** verify if water meter is given correct reading
- PC10.** disassemble and reassemble the water meter for verification or repair
- PC11.** take action as per standard operating procedure to troubleshoot common faults in water meters
- PC12.** clean the disassembled parts

Monitor and maintain flow meters

To be competent, the user/individual on the job must be able to:

- PC13.** check the range and zero setting of the flow meter
- PC14.** inspect for bearing wear out, deposits in flow meter or corrosion of attached pipes
- PC15.** take action as per standard operating procedure to troubleshoot common faults in flow meters

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Operate and maintain and repair pipelines

To be competent, the user/individual on the job must be able to:

- PC16.** operate water pipelines with positive pressure and by opening and shutting off the valves gradually
- PC17.** flush the system to clear sediments
- PC18.** service the valve chamber and valves
- PC19.** inspect the pipelines for damage, wear and tear, leakage, entrainment and water hammer
- PC20.** locate the leaks in the pipes
- PC21.** replace faulty parts like gaskets, valves joints and pipes that are not repairable
- PC22.** repair damaged pipelines
- PC23.** apply cement mortar lining on corroded pipes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to the task
- KU2.** workplace safety requirements, hazard reporting and handling procedures
- KU3.** overview of water distribution and storage systems at water supply stations
- KU4.** operational and maintenance requirements and procedures for sump and storage tanks
- KU5.** operational and maintenance requirements and procedures for water meters
- KU6.** operational and maintenance requirements and procedures for flow meters
- KU7.** operational and maintenance requirements and procedures for pipelines
- KU8.** pipelines repair procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** measure all dimensions used in plumbing in metric scale
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS5.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS6.** plan one's daily tasks to achieve maximum productivity
- GS7.** be punctual and work as per agreed priorities
- GS8.** listen to customer's concerns and doubts carefully and address them
- GS9.** be courteous and polite with customers and team members
- GS10.** establish workable solutions for problems in hand in consultation with others and record them



Qualification Pack

- GS11.** breakdown relevant work process into its constituent activities for ease of analysis
- GS12.** identify ways to increase productivity and reduce errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain sump and storage tanks</i>	6	9	2	1
PC1. dewater sump and tank	2	2	0.5	0.5
PC2. clean, disinfect and rinse the sump and tank	1	2	0.5	-
PC3. desilt the area and pump house.	1	2	0.5	-
PC4. perform leakage test and carry out rectification if needed with epoxy coating, cement concreting, painting etc.	2	3	0.5	0.5
<i>Maintain water meters</i>	8	13	3	1
PC5. clean the dirt box or strainer	1	1	-	-
PC6. replace gaskets upon its wear and tear	1	2	0.5	-
PC7. clean the chamber where meter is installed	1	1	-	-
PC8. prevent water seepage into the water meter	1	2	0.5	-
PC9. verify if water meter is given correct reading	1	1	-	0.5
PC10. disassemble and reassemble the water meter for verification or repair	1	3	1	-
PC11. take action as per standard operating procedure to troubleshoot common faults in water meters	1	3	1	0.5
PC12. clean the disassembled parts	1	-	-	-
<i>Monitor and maintain flow meters</i>	5	7	2	1
PC13. check the range and zero setting of the flow meter	1	1	0.5	-
PC14. inspect for bearing wear out, deposits in flow meter or corrosion of attached pipes	2	3	0.5	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. take action as per standard operating procedure to troubleshoot common faults in flow meters	2	3	1	0.5
<i>Operate and maintain and repair pipelines</i>	10	20	7	4
PC16. operate water pipelines with positive pressure and by opening and shutting off the valves gradually	1	2	1	0.5
PC17. flush the system to clear sediments	1	1	-	0.5
PC18. service the valve chamber and valves	1	2	1	0.5
PC19. inspect the pipelines for damage, wear and tear, leakage, entrainment and water hammer	2	3	1	1
PC20. locate the leaks in the pipes	1	3	1	0.5
PC21. replace faulty parts like gaskets, valves joints and pipes that are not repairable	1	3	1	1
PC22. repair damaged pipelines	2	3	1	-
PC23. apply cement mortar lining on corroded pipes	1	3	1	-
NOS Total	29	49	14	7

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0154
NOS Name	Maintain of Water Distribution and Storage Systems at Water Supply Stations
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

PSC/N0155: Operate and Maintain Water Treatment Units

Description

This unit deals in detail with the operation and maintenance of water treatment units, such as slow sand filters and chlorinators.

Scope

The scope covers the following :

- Operate and maintain water treatment units

Elements and Performance Criteria

Operate and maintain water treatment units

To be competent, the user/individual on the job must be able to:

- PC1.** start and shut off the filtration process
- PC2.** adjust the rate of filtration as needed
- PC3.** add chemicals, such as ammonia, chlorine, or lime, to disinfect water or other liquids
- PC4.** inspect equipment on a regular basis to ensure proper functioning and adherence to safety standards
- PC5.** monitor operating conditions, meters, and gauges
- PC6.** collect and test water and sewage samples
- PC7.** record meter and gauge readings, and operational data
- PC8.** operate equipment to purify and clarify water, or to process or dispose of sewage
- PC9.** clean and maintain equipment, tanks, filter beds, and other work areas

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards, policies, and procedures followed in the company relevant to the task
- KU2.** workplace safety requirements, hazard reporting and handling procedures
- KU3.** overview of water treatment units
- KU4.** commonly used water treatment units such as slow sand filters and chlorinators
- KU5.** operational and maintenance requirements and procedures for water treatment units
- KU6.** start and shut off the filtration process
- KU7.** rate of filtration and need for adjustment
- KU8.** chemicals used in the water treatment units
- KU9.** when and how to add chemicals, such as ammonia, chlorine, or lime, to disinfect water or other liquids
- KU10.** importance of regular inspection and the inspection process



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- KU11.** various meter and gauges and information to be gathered from them
- KU12.** importance and procedure of water and sewage testing
- KU13.** importance of maintaining records

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write common words/signs and set phrases used in the work
- GS2.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS3.** measure all dimensions used in plumbing in metric scale
- GS4.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS6.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS7.** spot discrepancies or errors and select the most efficient solution
- GS8.** plan one's daily tasks to achieve maximum productivity
- GS9.** be punctual and work as per agreed priorities
- GS10.** listen to customer's concerns and doubts carefully and address them
- GS11.** be courteous and polite in communications with customers and team
- GS12.** establish workable solutions for problems in hand in consultation with others and record them
- GS13.** identify ways to improve quality of work and reduce errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate and maintain water treatment units</i>	30	49	14	7
PC1. start and shut off the filtration process	3	4	1	-
PC2. adjust the rate of filtration as needed	3	5	1	1
PC3. add chemicals, such as ammonia, chlorine, or lime, to disinfect water or other liquids	3	6	1	1
PC4. inspect equipment on a regular basis to ensure proper functioning and adherence to safety standards	4	6	2	1
PC5. monitor operating conditions, meters, and gauges	3	6	2	1
PC6. collect and test water and sewage samples	3	7	2	1
PC7. record meter and gauge readings, and operational data	3	5	2	-
PC8. operate equipment to purify and clarify water, or to process or dispose of sewage	4	6	2	1
PC9. clean and maintain equipment, tanks, filter beds, and other work areas	4	4	1	1
NOS Total	30	49	14	7

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0155
NOS Name	Operate and Maintain Water Treatment Units
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	24/02/2022
Next Review Date	24/02/2025
NSQC Clearance Date	24/02/2022

Qualification Pack

PSC/N0136: Apply Health and Safety Practices at the Workplace

Description

This unit is about following safety procedures, communicating potential hazards and dangers of accidents on the job.

Scope

The scope covers the following :

- Follow safety measures to avoid accidents
- Daily inspection of tools and equipment
- Deal with emergencies

Elements and Performance Criteria

Follow safety measures to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC1.** identify risks and hazards at the workplace
- PC2.** wear personal protective equipment (PPE) as per the type of plumbing work
- PC3.** place protective barricades and signages around the pits and trenches
- PC4.** isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents
- PC5.** adhere to organisational procedures for reporting hazards and incidents to relevant authorities
- PC6.** establish ventilation before entering underground work areas
- PC7.** work safely in and around trenches, elevated places and confined areas
- PC8.** ensure tools and hazardous materials are not left unattended
- PC9.** ensure good housekeeping in order to prevent hazards e.g. fire
- PC10.** dispose waste materials and used PPE according to regulations and codes of practice

Follow hygiene and sanitation practices

To be competent, the user/individual on the job must be able to:

- PC11.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc.
- PC12.** clean and disinfect work area, materials/supplies, equipment etc. before and after use.
- PC13.** report hygiene and sanitation issues to appropriate authority

Use tools, equipment and materials safely

To be competent, the user/individual on the job must be able to:

- PC14.** check that the tools, equipment and materials are in good condition and as per industry standards before use
- PC15.** use power tools and machinery that are grounded
- PC16.** replace or repair split or loose tools before use
- PC17.** store and transport various plumbing materials safely

Qualification Pack

Deal with emergencies

To be competent, the user/individual on the job must be able to:

- PC18.** follow workplace emergency and evacuation procedures
- PC19.** use a fire extinguisher correctly
- PC20.** use safe methods to free a person from electrocution
- PC21.** administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** possible causes of risk, hazard or accident in the workplace
- KU2.** organisational procedures for upkeep of tools and plumbing materials, health and safety
- KU3.** location of all the general health and safety equipment in the workplace
- KU4.** meaning of hazards and risks
- KU5.** hazardous environment encountered during work such as underground areas, elevated areas, areas with water and electricity supply, presence of biological waste, under construction sites etc.
- KU6.** work practices and precautions to control and prevent risks, hazards and accidents
- KU7.** importance of each personal protective equipment used such as eye protection mask, hard hats, gloves, apron, rubber boots etc.
- KU8.** tools and plumbing equipment as per latest industry standards
- KU9.** preventative and remedial actions to be taken in case of exposure to toxic materials
- KU10.** specific safety and health related problems faced in domestic, commercial and institutional setups
- KU11.** various causes of fire and precautionary activities to prevent the fire accident
- KU12.** techniques of using the different fire extinguishers
- KU13.** rescue techniques applied during a fire hazard
- KU14.** various types of safety signs and meaning
- KU15.** appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, minor burns, poisoning, eye injuries etc.
- KU16.** potential injuries and ill health associated with incorrect handling of tools and equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write an accident/incident report in local language or English
- GS2.** read and comprehend labels, charts, signages, manuals, plumbing symbols etc.
- GS3.** question coworkers appropriately in order to clarify instructions and other issues
- GS4.** plan and organize the work schedule, work area, tools, equipment and materials for improved productivity



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- GS5.** determine key considerations and priorities when faced with problems
- GS6.** seek official and authorised sources of help and guidance to resolve problems that cannot be solved at one's level of authority

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow safety measures to avoid accidents</i>	13	32	-	3
PC1. identify risks and hazards at the workplace	2	2	-	0.5
PC2. wear personal protective equipment (PPE) as per the type of plumbing work	2	2	-	0.5
PC3. place protective barricades and signages around the pits and trenches	1	4	-	0.5
PC4. isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents	1	4	-	-
PC5. adhere to organisational procedures for reporting hazards and incidents to relevant authorities	2	4	-	0.5
PC6. establish ventilation before entering underground work areas	1	4	-	-
PC7. work safely in and around trenches, elevated places and confined areas	1	3	-	-
PC8. ensure tools and hazardous materials are not left unattended	1	3	-	-
PC9. ensure good housekeeping in order to prevent hazards e.g. fire	1	3	-	0.5
PC10. dispose waste materials and used PPE according to regulations and codes of practice	1	3	-	0.5
<i>Follow hygiene and sanitation practices</i>	3	8	-	1
PC11. follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc.	1	3	-	0.5
PC12. clean and disinfect work area, materials/supplies, equipment etc. before and after use.	1	3	-	0.5
PC13. report hygiene and sanitation issues to appropriate authority	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Use tools, equipment and materials safely</i>	6	9	-	1
PC14. check that the tools, equipment and materials are in good condition and as per industry standards before use	1	3	-	0.25
PC15. use power tools and machinery that are grounded	1	2	-	0.25
PC16. replace or repair split or loose tools before use	2	2	-	0.25
PC17. store and transport various plumbing materials safely	2	2	-	0.25
<i>Deal with emergencies</i>	8	14	-	2
PC18. follow workplace emergency and evacuation procedures	2	3	-	0.5
PC19. use a fire extinguisher correctly	2	4	-	0.5
PC20. use safe methods to free a person from electrocution	2	3	-	0.5
PC21. administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc.	2	4	-	0.5
NOS Total	30	63	-	7

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0136
NOS Name	Apply Health and Safety Practices at the Workplace
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing, Water Supply & Water Treatment & Quality Control, Sewerage & Sewage Treatment, Drainage, Water Harvesting & Ground Recharging, Firefighting & Safety Systems, Gas & Piping (Industrial / Medical), HVAC & Steam, Manufacturing of Plumbing / Firefighting Products, Micro-Hydel Power
Occupation	Plumbing Systems Installation and Maintenance, Designing, Sales, Material Handling, Plumbing Servicing, Plumbing Plant Operations
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0137: Work Effectively with Others

Description

This unit covers basic etiquette and competencies required in behavior and interactions with others at the workplace.

Scope

The scope covers the following :

- Communicate effectively
- Work in a team effectively
- Respect diversity

Elements and Performance Criteria

Communicate effectively

To be competent, the user/individual on the job must be able to:

- PC1.** obtain complete information and instructions
- PC2.** seek clarifications from appropriate source when required
- PC3.** provide information accurately and clearly
- PC4.** use inclusive language (verbal, non-verbal and written) that is gender, disability and culturally sensitive

Work in a team effectively

To be competent, the user/individual on the job must be able to:

- PC5.** prioritize tasks as required
- PC6.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC7.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict
- PC8.** act upon constructive feedback from others

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC9.** transact with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC10.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace
- PC11.** report incidents of harassment and discrimination to appropriate authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to employment, behavior, harassment, discrimination and performance conditions

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- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** escalation matrix and procedures for reporting work and employment related issues
- KU4.** types of harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU5.** importance of effective communication and the impact of poor communication on the employee, the employer and the customer
- KU6.** importance of teamwork in organizational and individual success
- KU7.** various components of effective communication such as tone and pitch
- KU8.** importance of ethics and discipline for professional success
- KU9.** how to express and address grievances appropriately and effectively
- KU10.** importance and ways of managing interpersonal conflict effectively
- KU11.** different types of disabilities and the challenges faced by persons with disability (PwD)
- KU12.** laws, acts and provisions defined for PwD by the statutory bodies
- KU13.** government and private schemes and benefits available for PwD
- KU14.** Importance of gender sensitivity and equality.
- KU15.** gender, disability and cultural biases, stereotypes and impact on others
- KU16.** gender and its concepts such as gender roles, gender spectrum, gender as an identity
- KU17.** legislations, grievance redressal mechanisms, and penalties against harassment in the workplace

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes, keep records, prepare to-do lists and take down instructions
- GS2.** write basic numbers, quantities and basic work-related terminology for operational requirements in the local language and English
- GS3.** read basic terminologies to accurately interpret labels, supervisor's instructions in the local language and English
- GS4.** read and interpret accurate information from work-related documents and various relevant work instructions and records in local
- GS5.** interact with the concerned personnel appropriately (correct protocol and manner of speaking etc.)
- GS6.** display active listening skills while interacting with co-workers and others in the workplace
- GS7.** deliver consistent and reliable service to internal and external customers
- GS8.** work with co-workers and supervisor to resolve any issues that threaten work quality as per the planned schedule

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively</i>	11	26	-	2
PC1. obtain complete information and instructions	3	8	-	-
PC2. seek clarifications from appropriate source when required	1	5	-	-
PC3. provide information accurately and clearly	3	7	-	1
PC4. use inclusive language (verbal, non-verbal and written) that is gender, disability and culturally sensitive	4	6	-	1
<i>Work in a team effectively</i>	11	25	-	3
PC5. prioritize tasks as required	3	8	-	1
PC6. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	8	-	1
PC7. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	4	-	1
PC8. act upon constructive feedback from others	2	5	-	-
<i>Respect diversity</i>	8	12	-	2
PC9. transact with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	3	5	-	1
PC10. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace	3	5	-	1
PC11. report incidents of harassment and discrimination to appropriate authority	2	2	-	-
NOS Total	30	63	-	7

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0137
NOS Name	Work Effectively with Others
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing, Water Supply & Water Treatment & Quality Control, Sewerage & Sewage Treatment, Drainage, Water Harvesting & Ground Recharging, Firefighting & Safety Systems, Gas & Piping (Industrial / Medical), HVAC & Steam, Manufacturing of Plumbing / Firefighting Products, Micro-Hydel Power
Occupation	Plumbing Systems Installation and Maintenance, Designing, Sales, Material Handling, Plumbing Servicing, Plumbing Plant Operations
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of total aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0151.Install Pump Systems and Related Equipment	30	49	14	7	100	14
PSC/N0152.Operate and Maintain Pumps and Related Machinery	30	49	14	7	100	14
PSC/N0153.Perform Maintenance of Local Ground Water Source and Water Quality Monitoring at Water Supply Stations	31	47	15	7	100	14
PSC/N0154.Maintain of Water Distribution and Storage Systems at Water Supply Stations	29	49	14	7	99	14
PSC/N0155.Operate and Maintain Water Treatment Units	30	49	14	7	100	14
PSC/N0136.Apply Health and Safety Practices at the Workplace	30	63	-	7	100	10
PSC/N0137.Work Effectively with Others	30	63	-	7	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	0	0	50	10



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	230	399	0	0	749	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.